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Influence of Fit-checking Material and Cleaning Protocols on the Bond Strength of Lithium Disilicate Glass-ceramics

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Clinical Relevance

Articulating spray and fit-checker liquid are viable options and should be encouraged for indirect restoration fit-check (instead of articulating paper) since these materials can be properly removed from a lithium disilicate surface before bonding procedures.

SUMMARY

Objectives: To evaluate the bond strength of a resin luting agent to a lithium disilicate glass-ceramic after the use of different fit-checking materials and cleaning protocols.

Methods and Materials: Two hundred and forty-two (242) ceramic specimens were etched with 5% hydrofluoric acid for 20 seconds and distributed into 22 groups (n=10), in total. Four (4) groups were created based on fit-checking material and that had no following cleaning protocol: no fit-checking material used (control group); articulating paper; articulating spray; and fit-checker liquid. For each fit-checking material (3), 6 cleaning protocols were tested creating an additional 18 groups (n=10): air/water spray; 70% alcohol (ethanol); acetone; 35% phosphoric acid; 5% hydrofluoric acid; and

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a commercially available cleaning paste (Ivoclean, Ivoclar Vivadent). Silane and bonding resin were applied to all ceramic surfaces. Resin luting agent cylinders (1 mm in diameter) were created using silicone matrices, light-cured, and specimens were stored in deionized water at 37°C for 24 hours. Microshear bond strength test (μSBS) was performed on a universal testing machine (DL 500, EMIC) at a crosshead speed of 1 mm/min until failure. Results were statistically analyzed using ANOVA and Tukey's test (α=0.05). Results: Articulating paper and fit-checker liquid, when not properly removed, negatively affected the bond strength (p<0.05). None of the tested cleaning protocols were effective for articulating paper (p<0.05). There was no significant difference in bond strength after the cleaning protocols between articulating spray and fit-checker liquid when compared to the control group (no contamination) (p>0.05).

Conclusion: The cleaning protocols tested can effectively restore the bond strength of resin luting agents to lithium disilicate ceramics that were exposed to articulating spray or fit-checker liquid. The use of articulating paper is not recommended for fit-checking indirect lithium disilicate restorations.

INTRODUCTION

Lithium disilicate glass-ceramics used in restorative dentistry are highly attractive to dentists and patients due to the combination of excellent mechanical, chemical, and esthetic properties, such as the ability to mimic optical characteristics of enamel and dentin; low thermal conductivity; high flexural strength, biocompatibility; wear resistance; and chemical stability.[1-5] Also, glass-ceramics can be micromechanically and chemically bonded to resin cements and dental substrates, which ensures long-lasting restorations.[3,6-8]

For optimal bonding performance to lithium disilicate glass ceramics, the surfaces are etched with hydrofluoric acid (5-10%) to create a rough surface by dissolving the glassy matrix, increasing surface area and surface energy, which increases the wetting by polymeric materials.[8-13] In addition, a silane coupling agent is applied to ensure chemical interaction between the silica and methacrylate groups of the resin luting agent or bonding resin.[14-16] The association of hydrofluoric acid etching followed by silane application is the most adopted intaglio surface treatment of glass ceramics since it results in high bond strength values.

During try-in procedures, indirect restorations (veneers, inlays, onlays, overlays, or full crowns) may not properly seat on the prepared tooth, requiring selective removal of any interferences. Slight reductions can be made on the tooth preparation to improve the fit. If the adaptation is not ideal, it may impair the proper seating during bonding procedures, which may lead to occlusal interferences, marginal staining, carious lesions, or even fracture of the restorative material. Thus, clinicians can use "fit-checking" materials (such as articulating paper, spray, or liquid) which are placed/applied onto the intaglio surface between the indirect restoration and prepared tooth, indicating the precise location of any high spots. However, this procedure may leave fit-checking material remnants on the intaglio surface, which needs to be removed for proper bonding procedures.

Previous studies have described the negative effects of saliva and human blood on the bond strength between glass-ceramics and resin luting agents, as well as specific protocols to remove these organic contaminants.[17-19] However, to the best of our knowledge, there is no information in the dental literature about protocols for cleaning fit-checking remnants from the intaglio surface of lithium disilicate glass-ceramics.

Therefore, the aim of the present *in vitro* study was to evaluate the microshear bond strength of a resin luting agent and bonding resin to a lithium disilicate glass-ceramic after the use of three different fit-checking materials and six different cleaning protocols. The null hypotheses tested were: (1) the use of fit-checking materials will not affect the bond strength; and (2) the bond strength after different cleaning protocols will be statistically similar to the control group (no contamination). Furthermore, ceramic surface topographies were assessed by scanning electron microscopy to qualitatively assess the different cleaning protocols.

METHODS AND MATERIALS

Preparation of Lithium Disilicate Glass-Ceramic Specimens

Two hundred and forty-two (242) lithium disilicate glass-ceramic specimens (10 mm diameter x 3 mm thickness) (IPS e.max Press, shade LTA2, Ivoclar Vivadent, Schaan, Liechtenstein) were fabricated according to the manufacturer's instructions. Ceramic specimens were embedded in acrylic resin using polyvinyl siloxane (PVS) molds (20 mm diameter x 20 mm height) (Figure 1A). Specimens were submitted to sequential finishing and polishing using silicon carbide abrasive papers (#150 and #600, 3M ESPE, St

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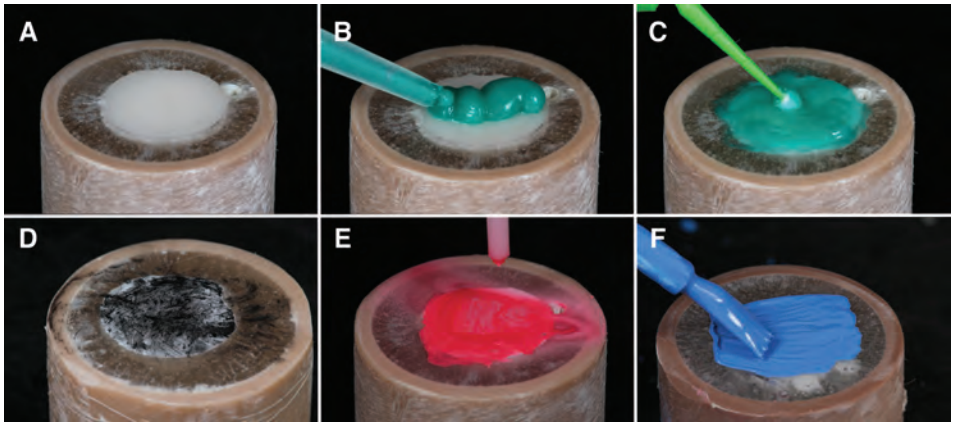


Figure 1. *Ceramic specimen treatment and fit-checking materials use. (A) Smooth flat surface of lithium disilicate glass-ceramic; (B) application of 5% hydrofluoric acid; (C) using a microbrush to spread the gel uniformly on the surface; (D) contamination with articulating paper; (E) contamination with articulating spray; and (F) contamination with fit-checker liquid.*

Paul, MN, USA) in a water-cooled automatic polisher (Metaserv 250, Buehler, Lake Buff, IL, USA) to obtain a flat, homogeneous, and smooth surface. All specimens were water-rinsed, immersed in an ultrasonic bath for 10 minutes, and air-dried using oil-free compressed air. Prepared ceramic surfaces were etched with 5% hydrofluoric acid (HF) (Potenza Attacco; PHS do Brasil, Joinville, SC, Brazil) for 20 seconds (Figure 1B and C), water-rinsed using oil-free air/water spray for 20 seconds and air-dried for 20 seconds. Materials used in this study are described in Table 1.

Ceramic Surface Contamination and Group Distributions

The etched ceramic specimens were randomly assigned into 22 groups (n=10) according to the type of fit-checking material: no fit-checking (control group), articulating paper, articulating spray, and fit-checker liquid (Figure 1D, E, and F). Six cleaning protocols (Table 2 and Figure 2) were tested for each type of fit-checking material (n=10): [AW] air/water spray; [70] 70% alcohol (ethanol); [AC] acetone; [PA] 35% phosphoric acid; [HF] 5% hydrofluoric acid; and [IV] a commercially available cleaning paste (Ivoclean, Ivoclar Vivadent). To evaluate the effect of fit-checking remnants at the bonding interface, materials were not removed from the ceramic surfaces (n=10) and compared to the control group. Ceramic surfaces were completely dried with oil-free compressed air. A silane coupling agent (Monobond N, Ivoclar Vivadent) was actively applied using a disposable brush for 15 seconds to previously etched ceramic surfaces, left to react for 60 seconds, and air-dried until solvents were completely evaporated. Then, a thin layer of a bonding resin (Bond, Scotchbond MultiPurpose, 3M ESPE) was applied to the ceramic surfaces and light-cured for 20 seconds using a polywave LED light curing unit (Bluephase N, Ivoclar Vivadent, 1200 mW/cm^2). To evaluate the effect of different cleaning protocols, the same procedures were performed after each cleaning protocol (described in detail in Table 2).

Microshear Bond Strength Test (μSBS)

For the μSBS test, 1-mm thick elastomeric molds (Oranwash L, Zhermack, Italy) were fabricated, each containing three (3) cylinder-shaped orifices (1 mm in diameter) and positioned over the bonding area. The orifices were filled with a light-curable resin luting agent (Variolink Esthetic, Shade Neutral, Ivoclar Vivadent) at room temperature. A Mylar strip and glass slab were placed over top with a vertical load of 250 g for 2 minutes to standardize resin luting agent cylinders. The vertical load and glass slab were removed, and resin luting agent cylinders were light-cured for 40 seconds at once using a polywave LED light curing unit (Bluephase N, 1200 mW/cm^2), with the curing tip in close contact to the Mylar strip. All specimens were stored in deionized water at 37°C for 24 hours. After storage time, elastomer molds were carefully sectioned using a #11 scalpel blade and removed. Cylinders were inspected for flaws or defects and discarded if any were detected. The μSBS assessment has been previously described.[8,11,19]

A stainless-steel orthodontic wire (diameter of 0.2 mm) was placed at the bonding interface and around each cylinder for μSBS. Tests were performed using a universal testing machine (EMIC DL 500, Emic, São José dos Pinhais, PR, Brazil) with a 100 N load cell at a crosshead speed of 1.0 mm/min until failure. Ten (10) specimens were tested for each group and the mean value of three resin luting agent cylinders was considered the mean μSBS (MPa) value of each specimen. Values were recorded and subjected to statistical analysis. Debonded areas were evaluated, and failure modes were classified as adhesive (mode 1); cohesive within resin luting agent (mode 2); cohesive within ceramic (mode 3); and mixed, involving resin luting agent, adhesive, and/or cohesive within ceramic (mode 4).

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Table 1: *Materials Used in This Study*

Material	Brand Name (Manufacturer)	Composition
Lithium disilicate glass-ceramic	IPS e.max Press (Ivoclar Vivadent, Schaan, Liechtenstein)	SiO_2, Li_2O, K_2O, P_2O5, ZrO_2, ZnO, other oxides, and ceramic pigments
Hydrofluoric acid	Potenza Attacco (PHS do Brasil)	5% hydrofluoric acid, pigments, thickeners, deionized water
Silane coupling agent	Monobond-N (Ivoclar Vivadent)	Alcohol solution of silane methacrylate, phosphoric acid methacrylate, and sulfide methacrylate
Phosphoric acid	Ultra-Etch (Ultradent Inc)	35% phosphoric acid, glycol, cobalt aluminate blue spinel
Alcohol	Álcool 70 Prolink (Prolink Indústria Química)	70% alcohol solution
Acetone	Lutex AP Eco (Cinco Produtos Químicos e Farmacêuticos Ltda, Colombo, PR, Brazil)	Acetone, alcohol, water, benzophenone-3, methylal, propylene glycol
Commercial cleaning paste	Ivoclean (Ivoclar Vivadent)	Sodium hydroxide, ZrO_2, water, polyethylene glycol, pigments
Articulating paper	AccuFilm II (Parkell Inc, Edgewood, NY, USA)	Solid material but the manufacturer has not informed the ingredients
Articulating spray	Arti-Spray (Bausch, Mashua, NH, USA)	Propane, butane, titanium dioxide
Fit-checker liquid	Super Filme (Kota Indústria e Comércio Ltda, São Paulo, SP, Brazil)	Liquid flask: Blue pigment, titanium dioxide, ethyl alcohol, isopropyl alcohol, mineral oil, ethylene glycol, and polysorbate 80 Solvent flask: ethyl alcohol, isopropyl alcohol, mineral oil, ethylene glycol, and polysorbate 80
Adhesive resin (bond)	Scotchbond MP (3M Oral Care, St Paul, MN, USA)	Bisphenol A diglycidyl dimethacrylate (BisGMA), 2-hydroxyethyl methacrylate (HEMA), amines, photoinitiator
Light-cured resin luting agent	Variolink Esthetic (Ivoclar Vivadent)	Urethane dimethacrylate (UDMA) and methacrylate monomers, ytterbium trifluoride and spheroid mixed oxide, initiators, stabilizers, pigments

Scanning Electron Microscopy Analysis (SEM)

Representative ceramic samples from each group (n=1) were submitted to SEM analysis to verify the presence, or not, of fit-checking material remnants after each cleaning protocol. Specimens were mounted on aluminum stubs and sputter-coated with gold (Bal-TEC - SCD 050; Balzers Union AG, Fürstentum, Liechtenstein) for 120 seconds at 40 mA. Surfaces were then observed in SEM (FEI Quanta 250, FEI Company, Eindhoven, The Netherlands) by the same operator using magnifications of 2000-7000× at 15-25 kV.

Statistical Analysis

Shapiro–Wilk analysis was performed to verify data normality. μSBS data of control groups were submitted to one-way ANOVA and Tukey's post-hoc test (α=0.05). The influence of different cleaning protocols was analyzed using one-way ANOVA and multiple comparisons were performed using Tukey's post hoc test (α=0.05).

RESULTS

Microshear Bond Strength Test (μSBS)

μSBS results are described in Tables 3 and 4. The control group without previous contamination yielded the highest bond strength values (17.8±4.9 MPa), statistically similar to the articulating spray not submitted to any cleaning method (16.1±2.8 MPa) (p>0.05). Articulating paper and fit-checker liquid

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Table 2: *Description of Groups and Cleaning Methods*

Group	Protocol
Control group	No fit-checking materials used
Articulating paper	Articulating paper was pressed and rubbed against ceramic surfaces for 5 s
Articulating spray	Articulating spray was applied on the ceramic surfaces from a distance of 10 mm for 5 s and allowed to dry for 20 s
Fit-checker liquid	Fit-checker liquid and solvent were mixed in a proportion of 1:1 for 5 s and actively applied on the ceramic surfaces for 5 s and allowed to dry for 20 s
Cleaning Methods for Ceramic Surfaces Contaminated with Fit-Checking Materials	
Air/water spray	Oil-free air/water spray applied at a distance of 10 mm for 20 s
70% alcohol	One drop of 70% alcohol was actively rubbed on the ceramic surfaces for 10 s using a disposable microbrush, followed by oil-free air/water spray at a distance of 10 mm for 10 s
Acetone	One drop was actively rubbed on the ceramic surfaces for 10 s using a disposable microbrush, followed by oil-free air/water spray at a distance of 10 mm for 10 s
Phosphoric acid	35% phosphoric acid was actively rubbed on the ceramic surfaces for 10 s using a disposable microbrush, followed by oil-free air/water spray at a distance of 10 mm for 10 s
Hydrofluoric acid	5% hydrofluoric acid was actively rubbed on the ceramic surfaces for 10 s using a disposable microbrush, followed by oil-free air/water spray at a distance of 10 mm for 10 s
Ivoclean	Commercial cleaning paste was actively rubbed on the ceramic surfaces for 10 s using a disposable microbrush, followed by oil-free air/water spray at a distance of 10 mm for 10 s

resulted in lower bond strength results when compared to the control group ($p<0.05$). Articulating paper resulted in the lowest bond strength values ($p<0.05$) (Table 3). None of the cleaning protocols used for the articulating paper were effective in restoring or improving bond strength when compared to the control group ($p<0.05$) (Table 4). All cleaning protocols resulted in statistically similar μSBS results ($p>0.05$). No statistical differences were found among cleaning methods for articulating spray ($p>0.05$). All methods resulted in statistically similar μSBS values when compared to the control group ($p>0.05$) (Table 4). For fit-checker liquid, no statistical differences were observed among cleaning protocols ($p>0.05$), except for acetone which resulted in the lowest bond strength ($p<0.05$) statistically similar to the group where the liquid was not removed before bonding procedures ($p>0.05$) (Table 4).

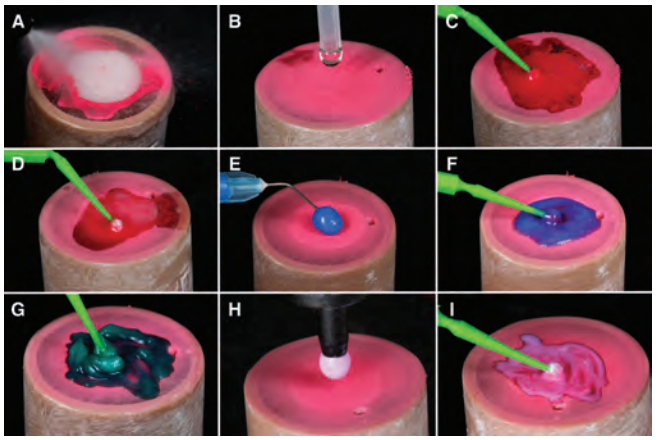


Figure 2. *Different cleaning protocols. (A) Air/water spray; (B and C) alcohol 70%; (D) acetone; (E and F) 35% phosphoric acid; (G) 5% hydrofluoric acid; and (H and I) cleaning paste (Ivoclean).*

Scanning Electron Microscope (SEM) Analysis

Scanning electron micrographs are presented in Figures 3, 4, 5, and 6. Uncontaminated surfaces etched with 5% HF depicted glassy matrix removal and exposure

Table 3: *Means and Standard Deviation of μSBS (MPa) for Control Group and Different Fit-Checking Materials*[a]

Groups	μSBS (MPa)
Control	17.8 (4.9) A
Articulating paper	7.4 (1.7) C
Articulating spray	16.1 (2.8) AB
Fit-checker liquid	13.4 (3.2) B
p-value	<0.001*

[a]*Different letters represent significant differences among groups (p<0.05). Abbreviations: MPa, megapascals; μSBS, microshear bond strength.*
**There was statistical difference among groups, including the control group.*

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Table 4: *Means and Standard Deviation of μSBS (MPa) for Control Group, Different Fit-Checking Materials, and Cleaning Methods*[a]

Cleaning Method	Articulating Paper	Articulating Spray	Fit-checker Liquid
No cleaning method used	7.4 (1.7) B	16.1 (2.8) A	13.4 (4.0) B
Air/water spray	7.4 (2.5) B	19.0 (3.3) A	17.2 (2.8) AB
70% alcohol	7.5 (3.1) B	17.8 (3.4) A	15.4 (5.0) AB
Acetone	8.2 (1.3) B	16.0 (4.3) A	13.6 (3.4) B
35% phosphoric acid	7.9 (1.8) B	17.9 (3.8) A	18.0 (4.8) A
5% hydrofluoric acid	9.8 (2.9) B	16.0 (3.9) A	14.6 (2.9) AB
Ivoclean	7.5 (1.6) B	18.1 (6.0) A	16.4 (3.3) AB
p-value	<0.001*	0.665	0.046*
Control group	17.8 (4.9) A		

[a]*Different letters within columns represent significant difference among groups (p<0.05). Abbreviations: MPa, megapascals; μSBS, microshear bond strength.*

**There was statistical difference among groups, including the control group.*

of lithium disilicate crystals (Figure 3A). Fit-checking material remnants were present on all surfaces that were not subjected to cleaning procedures (Figures 3B, 3C, and 3D) making it impossible to visualize lithium disilicate crystals. Cleaning methods were not able to properly remove articulating paper remnants from the etched ceramic surfaces (Figures 4 A-F). On the other hand, cleaning methods were able to remove articulating spray and fit-checker liquid remnants (Figures 5 and 6) as it was possible to observe the lithium disilicate crystals. Failure modes observed were 98.3% adhesive and 1.7% mixed failures at the bonded interface. No cohesive failures were observed.

DISCUSSION

For this *in vitro* study, the effect of different cleaning protocols of three fit-checking materials on the bond strength of a resin luting agent to a lithium disilicate glass-ceramic was investigated. The first null hypothesis was partially rejected since the use of articulating spray did not affect the bond strength to lithium disilicate. However, the use of articulating paper or fit-checker liquid had negative effects on bond strength. Furthermore, the second null hypothesis was also partially rejected, since all cleaning protocols were able to restore the bond strength when fit-checker liquid

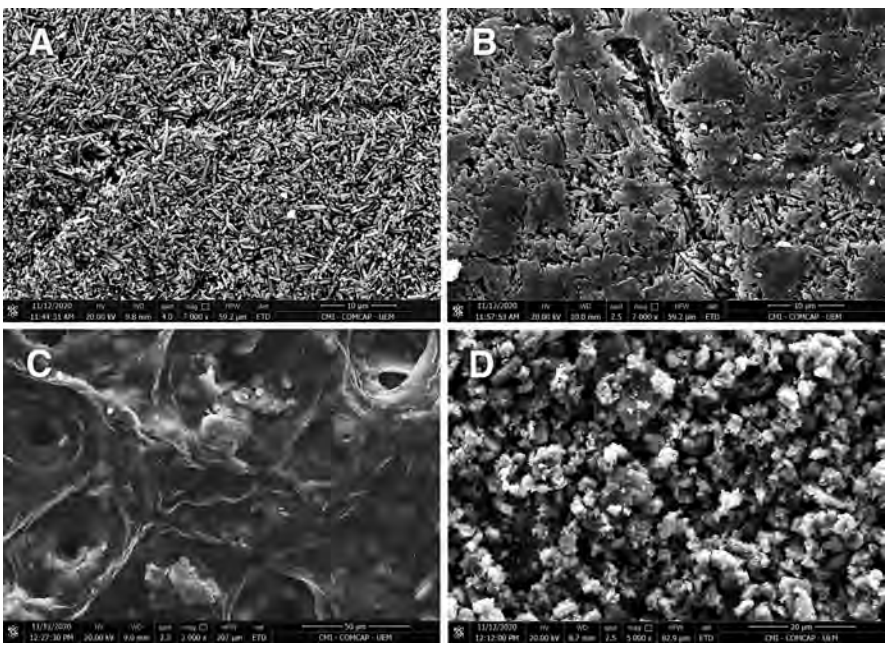


Figure 3. *Representative SEM images of evaluated groups. (A) Control group (not contaminated); (B) contaminated with articulating paper; (C) contaminated with articulating spray; and (D) contaminated with fit-checker liquid.*

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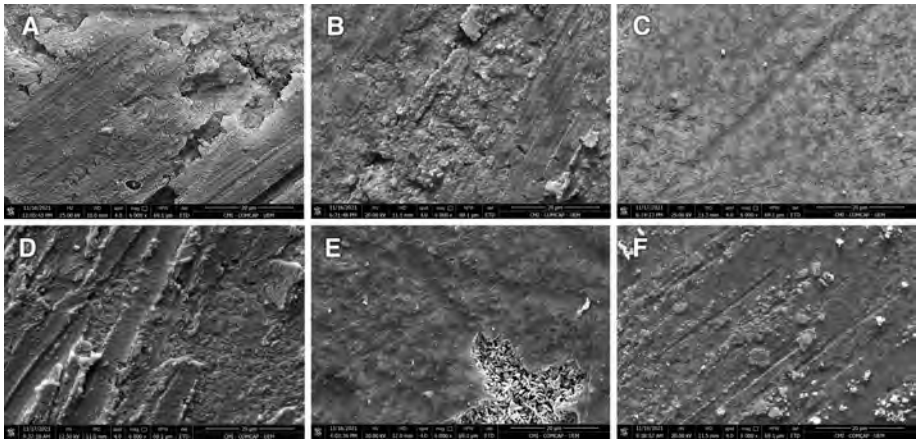


Figure 4. *Representative SEM images of ceramics contaminated with articulating paper after different cleaning protocols. (A) Air/water spray; (B) alcohol 70%; (C) acetone; (D) 35% phosphoric acid;(E) 5% hydrofluoric acid; and (F) cleaning paste (Ivoclean).*

was used, except when using acetone. Thus, the use of articulating spray and fit-checker liquid, after most cleaning protocols, seemed not to negatively interfere with the bond strength of resin luting agents to lithium disilicate ceramics. For the articulating paper, it was clear that remnants were not fully removed, which was confirmed by SEM.

Fit-checking materials are useful tools in dental practice since they can precisely indicate the adaptation of indirect restorations to prepared teeth surfaces. For this study, three different commercially available fit-checking materials were evaluated: articulating paper, articulating spray, and fit-checker liquid. After using all these materials, traces were clearly noticeable on the ceramic surfaces (Figure 1) which should be removed before bonding procedures. Although it was not possible to find laboratory studies that assessed the effects of remnants or different cleaning protocols for fit-checking

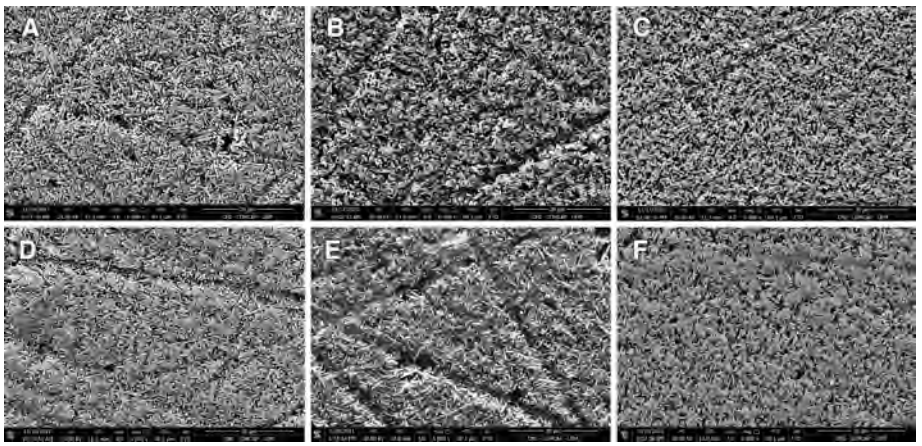


Figure 5. *Representative SEM images of ceramics contaminated with articulating spray after different cleaning protocols. (A) Air/water spray; (B) alcohol 70%; (C) acetone; (D) 35% phosphoric acid; (E) 5% hydrofluoric acid; and (F) cleaning paste (Ivoclean).*

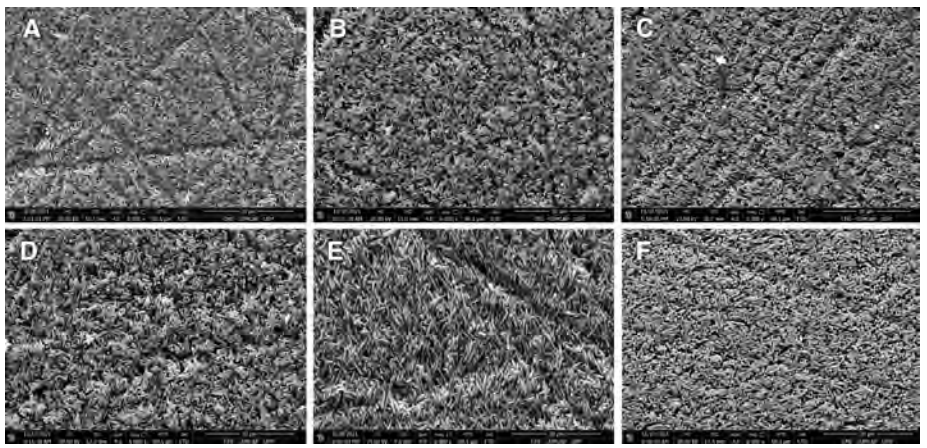


Figure 6. *Representative SEM images of ceramics contaminated with fit-checker liquid after different cleaning protocols. (A) Air/water spray; (B) alcohol 70%; (C) acetone; (D) 35% phosphoric acid; (E) 5% hydrofluoric acid; and (F) cleaning paste (Ivoclean).*

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materials, previous studies[17,19,20] reported a negative effect of other contaminants (saliva, human blood, or dental stone) on the bonding interface between glass ceramics and resin luting agents. Thus, it could be hypothesized that fit-checking material remnants would impair the ability of adequate mechanical and chemical interaction between the ceramic surface and bonding resin, and/or resin luting agent (Figures 3B and 3D).

Surprisingly, the use of articulating spray did not affect the bond strength to lithium disilicate ceramics (Table 3). Figure 3 demonstrates a complete obliteration of lithium disilicate crystals which, in theory, should have reduced the mechanical and chemical interaction between bonding resin or resin luting agent and the ceramic surface. After its application and solvent evaporation (propane and butane), probably only titanium dioxide particles (small and light in weight) remained on the ceramic surface. It may be suggested that the active application of a solvent-containing silane solution may have interacted with, or incorporated the titanium dioxide, dislodging these small particles which allowed an adequate chemical and mechanical interaction between bonding materials and ceramic surface. A similar effect could have happened during the active application of the bonding resin. These results should be interpreted with caution since the presence of a contaminant at the bonding interface should not be encouraged.

None of the proposed cleaning protocols were effective for the articulating paper (Figure 4). The use of air/water spray, 70% alcohol, acetone, 35% phosphoric acid, 5% hydrofluoric acid, or a commercial cleaning paste (Ivoclean) failed to dissolve and remove the remnants. Silane-coupling agent and bonding resin material were not able to interact with or incorporate the remnants of the articulating paper during their application, resulting in a significant reduction of bond strength when compared to the control group (Table 4). Therefore, it can be suggested that clinicians should not use articulating paper to fit-check ceramic restorations as it is not possible to be removed from etched lithium disilicate surfaces.

On the other hand, all described cleaning protocols were effective in removing articulating spray and fit-checker liquid remnants from ceramic surfaces (Table 4 and Figures 5 and 6), except acetone for the fit-checker liquid, resulting in similar bond strength when compared to control. The small titanium dioxide particles were probably easily removed by the different cleaning protocols since many of them used solvents. Likewise, the fit-checker liquid was easily solubilized and removed from the ceramic surfaces since this material contained alcohol as the solvent in the composition. Both articulating spray and fit-checker liquid are viable options for assessing the fit of glass ceramic restorations due to the easy cleaning, even by the simplest and least costly method (air/water spray).

Clinicians should note that the use of fit-checking materials leaves marks on both ceramic intaglio and tooth preparation surfaces. In order not to modify or weaken the glass-ceramic material, any slight interference that hampers satisfactory seating must be selectively removed from the prepared tooth. It is also necessary to emphasize that the evaluated fit-checking materials have high-contrasting colors to facilitate visualization. Thus, cleaning is important not only for bond strength but also for optimal esthetic outcomes, especially for thin glass-ceramic restorations (veneers, for example). Moreover, fit-checking material remnants at the margin of ceramic restorations may facilitate marginal staining, sensitivity, bacterial infiltration, and consequently carious lesions. Thus, clinicians should carefully remove fit-checking materials until no longer clinically visible. Future *in vitro* studies should be conducted to assess the effects of water storage and thermocycling on bond strength, as well as the use of different cleaning protocols.

CONCLUSIONS

Based on the results of this study, it can be concluded that articulating paper should not be used to fit-check lithium disilicate glass-ceramic restorations since its remnants cannot be removed from the etched ceramic surfaces before bonding procedures. Clinicians can use either articulating spray or fit-checker liquid since the proposed cleaning protocols were effective, resulting in bond strength similar to the uncontaminated control group.

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Conflict of Interest

The authors of this article certify that they have no proprietary, financial, or other personal interest of any nature or kind in any product, service, and/or company that is presented in this article.

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